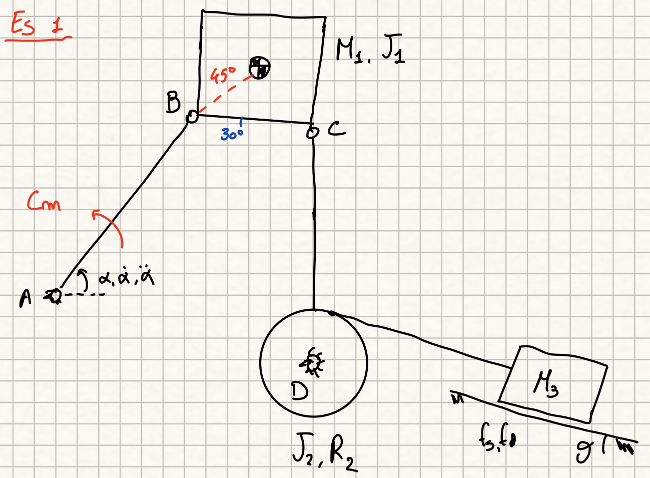
Es 1



$\overline{AB} = 1\ m$ $\quad M_1 = 1.5\ kg$ $\quad J_1 = 0.2\ kg\,m^2$

$\overline{BC} = 0.5m$ $\quad J_2 = 0.05\ kg\,m^2$

$\overline{CD} = 1.3\ m$ $\quad M_3 = 2\ kg$

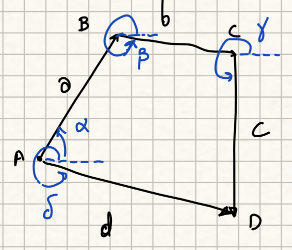
$R_2 = 0.5\ m$ $\quad f_d = 0.7$

$\vartheta = 30°$

$\alpha = 50°$

$\dot\alpha = 1\ rad/s$

$\ddot\alpha = 1\ rad/s^2$



Posizioni:

$$\begin{cases} a\cos\alpha + b\cos\beta + c\cos\gamma = d\cos\delta \\ a\sin\alpha + b\sin\beta + c\sin\gamma = d\sin\delta \end{cases}$$

Velocità:

$$\begin{cases} -a\dot\alpha\sin\alpha - b\dot\beta\sin\beta - c\dot\gamma\sin\gamma = 0 \\ a\dot\alpha\cos\alpha + b\dot\beta\cos\beta + c\dot\gamma\cos\gamma = 0 \end{cases}$$

$$\begin{bmatrix} b\sin\beta & c\sin\gamma \\ -b\cos\beta & -c\cos\gamma \end{bmatrix} \begin{Bmatrix} \dot\beta \\ \dot\gamma \end{Bmatrix} = \begin{Bmatrix} -a\dot\alpha\sin\alpha \\ a\dot\alpha\cos\alpha \end{Bmatrix} \Rightarrow \begin{aligned} \dot\beta &= -1.4845\ rad/s \\ \dot\gamma &= 0.8747\ rad/s \end{aligned}$$

Accelerazioni:

$$\begin{cases} -a\ddot\alpha\sin\alpha - a\dot\alpha^2\cos\alpha - b\ddot\beta\sin\beta - b\dot\beta^2\cos\beta - c\ddot\gamma\sin\gamma - c\dot\gamma^2\cos\gamma = 0 \\ a\ddot\alpha\cos\alpha - a\dot\alpha^2\sin\alpha + b\ddot\beta\cos\beta - b\dot\beta^2\sin\beta + c\ddot\gamma\cos\gamma - c\dot\gamma^2\sin\gamma = 0 \end{cases}$$

$$\begin{bmatrix} b\sin\beta & c\sin\gamma \\ -b\cos\beta & -c\cos\gamma \end{bmatrix} \begin{Bmatrix} \ddot\beta \\ \ddot\gamma \end{Bmatrix} = \begin{Bmatrix} -a\ddot\alpha\sin\alpha - a\dot\alpha^2\cos\alpha - b\dot\beta^2\cos\beta - c\dot\gamma^2\cos\gamma \\ a\ddot\alpha\cos\alpha - a\dot\alpha^2\sin\alpha - b\dot\beta^2\sin\beta - c\dot\gamma^2\sin\gamma \end{Bmatrix} = 0 \Rightarrow \begin{aligned} \ddot\beta &= -3.2848\ \frac{rad}{s^2} \\ \ddot\gamma &= 2.4494\ \frac{rad}{s^2} \end{aligned}$$

$$\vec{v}_B = \dot{\alpha}\vec{k} \wedge a(\cos\alpha\,\vec{i} + \sin\alpha\,\vec{j}) = \dot{\alpha} a(\cos\alpha\,\vec{j} - \sin\alpha\,\vec{i}) = (-0.776\,\vec{i} + 0.6428\,\vec{j})\ m/s$$

$$\vec{v}_G = \vec{v}_B + \dot{\beta}\vec{k} \wedge \frac{\sqrt{2}b}{2}\left(\cos(\beta+45)\vec{i} + \sin(\beta+45)\vec{j}\right) \qquad |v_B| = 1\ m/s$$

$$= \vec{v}_B + \dot{\beta}\frac{\sqrt{2}}{2}b\left(\cos(\beta+45°)\vec{j} - \sin(\beta+45)\vec{i}\right) = (-0.6302\,\vec{i} + 0.1358\,\vec{j})\ m/s$$

$$|v_G| = 0.6447\ m/s$$

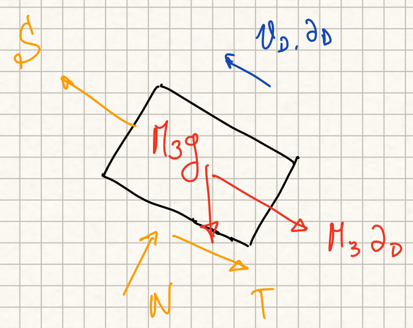
$$\vec{a}_B = \ddot{\alpha} a(\cos\alpha\,\vec{j} - \sin\alpha\,\vec{i}) - \dot{\alpha}^2 a(\cos\alpha\,\vec{i} + \sin\alpha\,\vec{j})$$

$$\vec{a}_G = \vec{a}_B + \ddot{\beta}\frac{\sqrt{2}}{2}b\left(\cos(\beta+45°)\vec{j} - \sin(\beta+45)\vec{i}\right) - \dot{\beta}^2\frac{\sqrt{2}}{2}b\left(\cos(\beta+45)\vec{i} + \sin(\beta+45)\vec{j}\right) =$$

$$= (-1.860\,\vec{i} - 1.4467\,\vec{j})\ m/s^2 \quad \Rightarrow \quad |a_G| = 2.357\ m/s^2$$

$$\frac{dE_c}{dt} = M_1 a_G v_G + J_1\dot{\beta}\ddot{\beta} + J_2\dot{\gamma}\ddot{\gamma} + M_3\dot{\gamma}\ddot{\gamma}R_2^2 = 3.6179$$

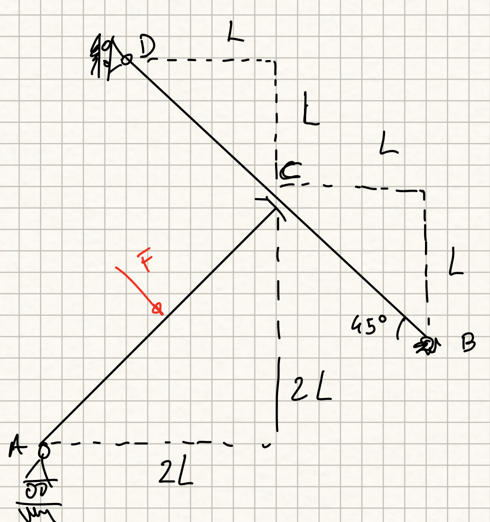
$$\Sigma W_{ATT} = -M_1 g\, v_{Gy} - M_3 g\,\dot{\gamma}R_2\sin\vartheta - M_3 g\cos\vartheta\, f_d\,\dot{\gamma}R_2 + C_m\dot{\alpha} \Rightarrow C_m = 15.1096\ Nm$$



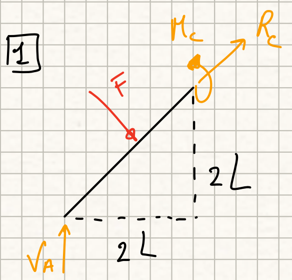
$$\begin{cases} N = M_3 g\cos\vartheta \\ T = f_d M_3 g\cos\vartheta \\ S = M_3 a_D + T + M_3 g\sin\alpha = M_3 a_D + f_d M_3 g\cos\vartheta + M_3 g\sin\vartheta \end{cases}$$

$$S = 24.15\ N$$

Es 2

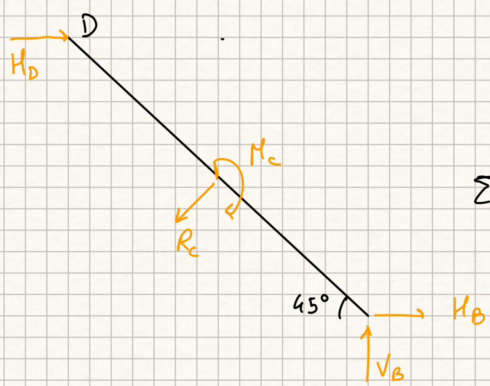


1



$$\begin{cases} R_c \frac{\sqrt{2}}{2} + F \frac{\sqrt{2}}{2} = 0 \Rightarrow R_c = -F \\ R_c \frac{\sqrt{2}}{2} + V_A - F \frac{\sqrt{2}}{2} = 0 \Rightarrow V_A = F\sqrt{2} \\ \Sigma M_A \quad M_c - F \frac{2\sqrt{2}L}{2} = 0 \Rightarrow M_c = \sqrt{2}FL \end{cases}$$

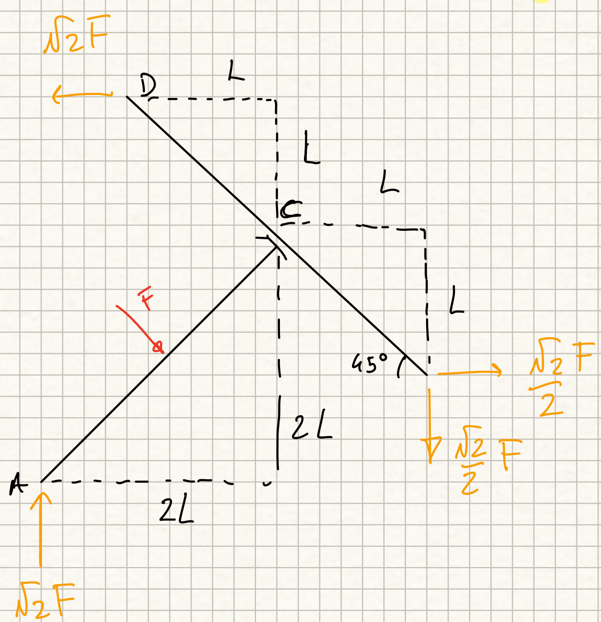
2



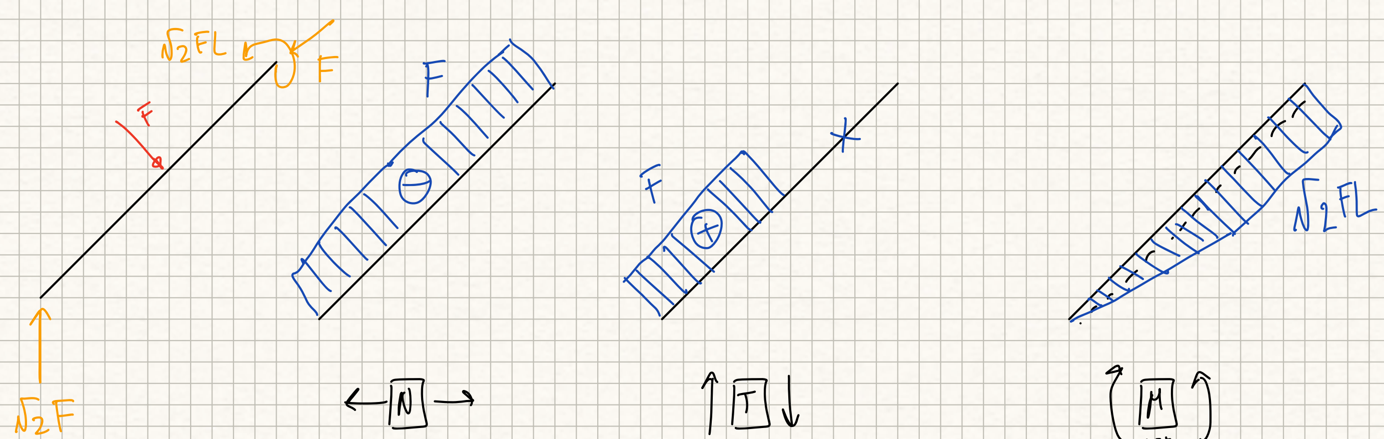
$$\begin{cases} H_D + H_B - R_c \frac{\sqrt{2}}{2} = 0 \\ V_B = R_c \frac{\sqrt{2}}{2} \Rightarrow V_B = -\frac{\sqrt{2}}{2} F \\ \Sigma M_B \quad -H_D \frac{\sqrt{2}}{2} \cdot 2\sqrt{2}L - M_c + R_c \sqrt{2} L = 0 \end{cases}$$

$$H_D = \frac{-F\sqrt{2}L - \sqrt{2}FL}{2L} = -\sqrt{2}F$$

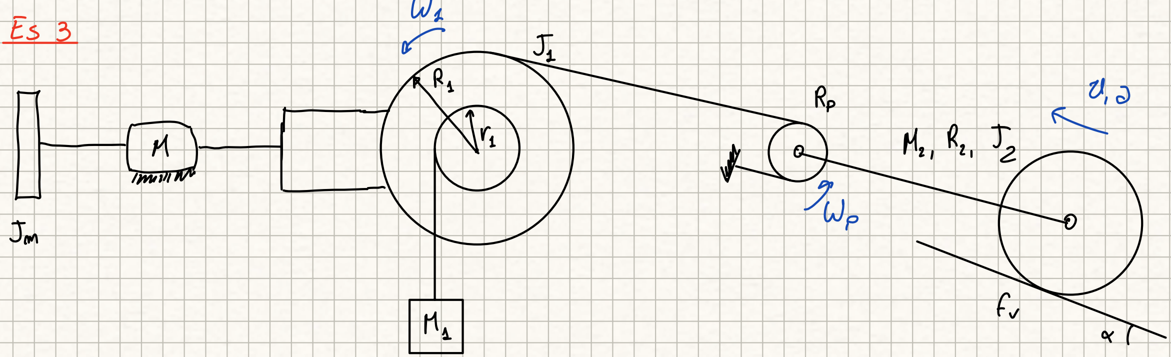
$$H_B = R_c \frac{\sqrt{2}}{2} + \sqrt{2}F = -F\frac{\sqrt{2}}{2} + \sqrt{2}F = \frac{\sqrt{2}}{2}F$$



AZIONI ASTA AC :



Es 3



$\tau = \frac{1}{10}$, $\eta_D = 0.8$, $\eta_R = 0.7$

$f_v = 0.05$, $f_s = 0.7$

$\alpha = 10°$

$R_1 = 0.7 m$, $r_1 = 0.1 m$

$R_2 = 0.5 m$, $R_p = 0.1 m$

$M_1 = 3 kg$ $M_2 = 30 kg$

$J_m = 0.5 kgm^2$; $J_1 = 0.05 kgm^2$

$J_2 = 0.3 kg m^2$

$C_{m0} = 30 Nm$

MOTORE:

$$W_1 = C_m \omega_m - J_m \omega_m \dot{\omega}_m$$

UTILIZZATORE

$$\omega_1 = \tau \omega_m , \quad \omega_p = \tau \omega_m \frac{R_1}{2R_p} , \quad v = \tau \omega_m \frac{R_1}{2} , \quad \omega_2 = \tau \omega_m \frac{R_1}{2R_2}$$

$$\Sigma W_{ATT} = -M_2 g \tau \omega_m \frac{R_1}{2} \sin\alpha + M_1 g r_1 \tau \omega_m - M_2 g \cos\alpha f_v \tau \omega_m \frac{R_1}{2} =$$

$$= \omega_m \tau \left[-M_2 g \frac{R_1}{2} \sin\alpha + M_1 g r_1 - M_2 g \cos\alpha f_v \frac{R_1}{2} \right] = \underbrace{C_u^*}_{-2.0016} \omega_m$$

$$\frac{dE_c}{dt} = \left[M_1 \tau^2 r_1^2 + J_1 \tau^2 + M_2 \tau^2 \frac{R_1^2}{4} + J_2 \tau^2 \frac{R_1^2}{4R_2^2} \right] \omega_m \dot{\omega}_m = \underbrace{J_u^*}_{0.0390} \omega_m \dot{\omega}_m$$

$$W_2 = C_u^* \omega_m - J_u^* \omega_m \dot{\omega}_m$$

1) C_m a regime: $W_2 < 0 \Rightarrow$ MOTO DIRETTO

$$\eta_D C_m \omega_m + C_u^* \omega_m = 0 \Rightarrow C_m = 2.502 \text{ Nm}$$

2) accelerazione allo spunto: IPOTIZZO DIRETTO

$$\eta_D \left[C_{m0} - J_m \dot{\omega}_m \right] + C_u^* - J_u^* \dot{\omega}_m = 0$$

$$\dot{\omega}_m = \frac{\eta_D C_{m0} + C_u^*}{\eta_D J_m + J_u^*} = 50.10 \text{ rad}/s^2 \Rightarrow a = 1.7538 \text{ m}/s^2$$

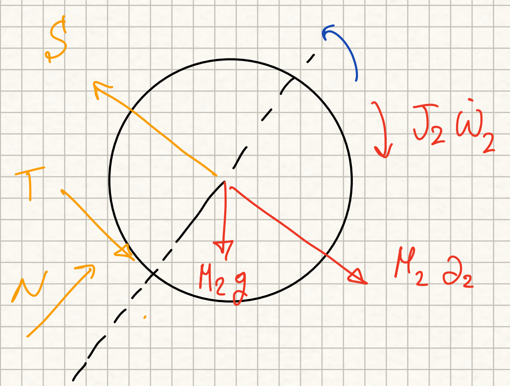
verifica moto: $W_1 = \omega_m \left[C_{m0} - J_m \dot{\omega}_m \right] = 4.94 \omega_m$ OK

3) C_m t.c. $a = 1.5 m/s^2$

$$\dot{\omega}_m = \frac{2a}{\tau R_1} = 42.857 \frac{rad}{s^2} \Rightarrow W_2 < 0 \Rightarrow \text{diretto}$$

$$\eta_D C_m - \eta_D J_m \dot{\omega}_m + C_u^* - J_u^* \dot{\omega}_m = 0 \Rightarrow C_m = \frac{\eta_D J_m \dot{\omega}_m - C_u^* + J_u^* \dot{\omega}_m}{\eta_D} = 26.0209 \ Nm$$

a) aderenza da (3)



$$\begin{cases} N = M_2 g \cos\alpha \\ J_2 \dot{\omega}_2 + M_2 g \cos\alpha \, f_v R_2 = T R_2 \end{cases}$$

$$T = \frac{J_2 \dot{\omega}_m}{R_2} + M_2 g \cos\alpha \, f_v = 16.2914 \ N$$

$$T_{lim} = M_2 g \cos\alpha \, f_s = 202.88 N \Rightarrow \text{OK}$$